

SENTORA

A Privacy-First, Voice-First AI Journaling Application

CAPTURE your thoughts naturally



UNDERSTAND them using AI



REDISCOVER meaningful memories over time

Project Deep-Dive & Pitch Document

Concept • Vision • Problem Statement • Architecture • Technology • Workflow •
Use Cases • Privacy • Future Scope

Document Overview

- **1. Project at a Glance** – what Sentora is and the one-line pitch.
- **2. Vision and Core Philosophy** – the long-term purpose behind the system.
- **3. Problems Being Solved** – why traditional journaling is not enough.
- **4. Complete User Journey** – from voice recording to saved memory.
- **5. System Architecture** – how frontend, backend, AI, and storage interact.
- **6. Technology Stack** – every major technology and its role.
- **7. AI Analysis and Smart Search** – how unstructured thoughts become organized memories.
- **8. Privacy Approach** – what stays local and what is sent for AI analysis.
- **9. Real-World Use Cases** – practical scenarios where Sentora is useful.
- **10. Pitch and Viva Preparation** – concise explanations for presentations.
- **11. Future Scope** – how the prototype can evolve.

1. Project at a Glance

Sentora is a privacy-first, voice-first AI journaling application that allows users to speak their thoughts naturally, converts those thoughts into structured journal entries, uses AI to understand and organize them, and helps users rediscover meaningful memories through intelligent search and personal insights.

Sentora is not merely a voice recorder, a notes application, or an AI chatbot. Its value comes from combining multiple capabilities into one personal memory experience.

VOICE + JOURNALING + AI UNDERSTANDING + MEMORY RETRIEVAL



SENTORA: A searchable and organized personal memory system

Core proposition: Traditional journals collect information. Sentora aims to capture, understand, organize, and help users rediscover information.

2. Vision and Core Philosophy

The long-term vision of Sentora is to make personal reflection as natural as speaking while making years of personal memories easier to organize and rediscover.

As journal entries accumulate, users often remember the meaning of a past experience but not the exact date or wording. For example, a user may remember being stressed about a project deadline but not remember which entry contained that experience.

Traditional Journal

A collection of separate documents



Sentora

A searchable personal memory system

- **Voice-first:** Reduce the friction of typing long entries.
- **AI-assisted:** Convert unstructured thoughts into useful structured information.
- **Local-first:** Keep saved journal history primarily on the user's device.
- **Memory-oriented:** Help users return to meaningful moments later.

3. Problems Sentora Solves

Problem 1: Journaling requires effort. Typing can feel slow and tiring, particularly after a busy day. Many people can comfortably speak for several minutes but may avoid writing the same amount of information.



Problem 2: Journal entries become difficult to organize. Over time, a chronological list of entries can become difficult to navigate. Sentora enriches each entry with additional metadata.

- Summary
- Emotional tone / mood classification
- Topics
- Keywords
- Date and journal activity information

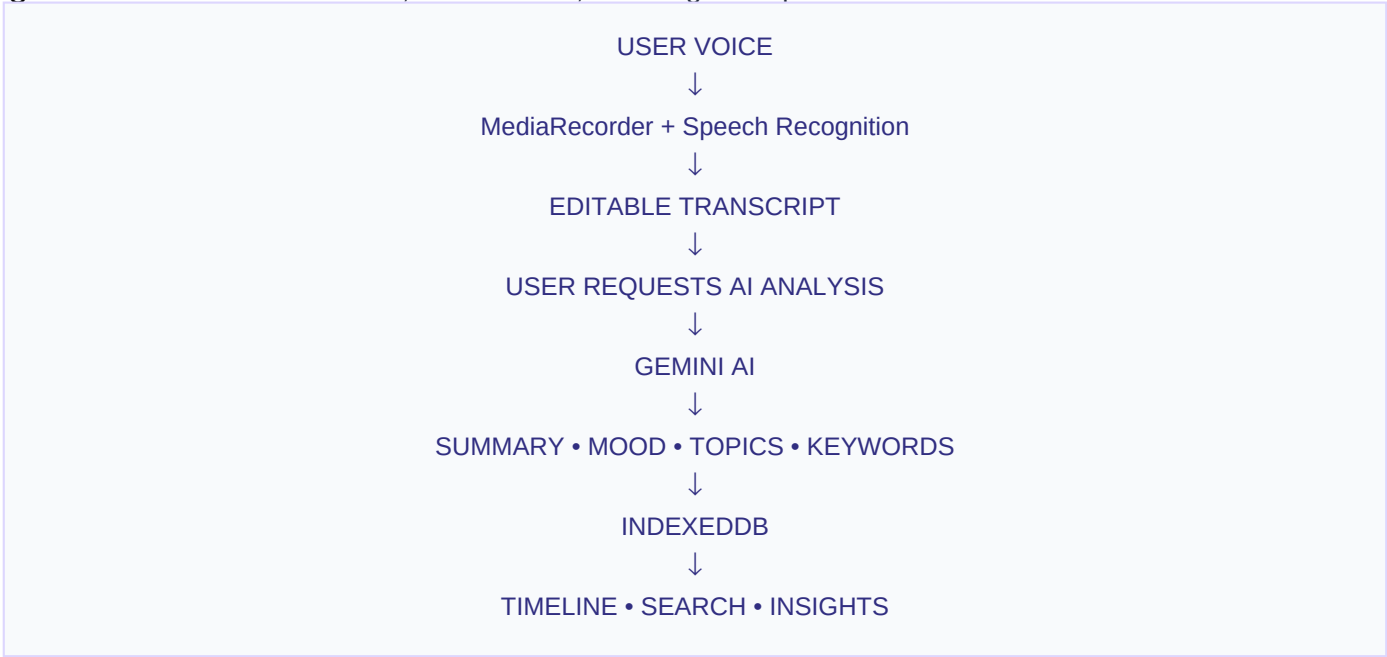
Problem 3: Searching memories is difficult. Users may remember an idea or feeling without remembering the exact words they used. Sentora searches across multiple representations of an entry rather than only a single title or exact keyword.

Problem 4: Personal patterns are hard to notice manually. Mood-related language, frequently discussed topics, and journaling activity may become easier to observe when visualized over time. These insights are reflective, not medical or psychological diagnoses.

4. Complete User Journey

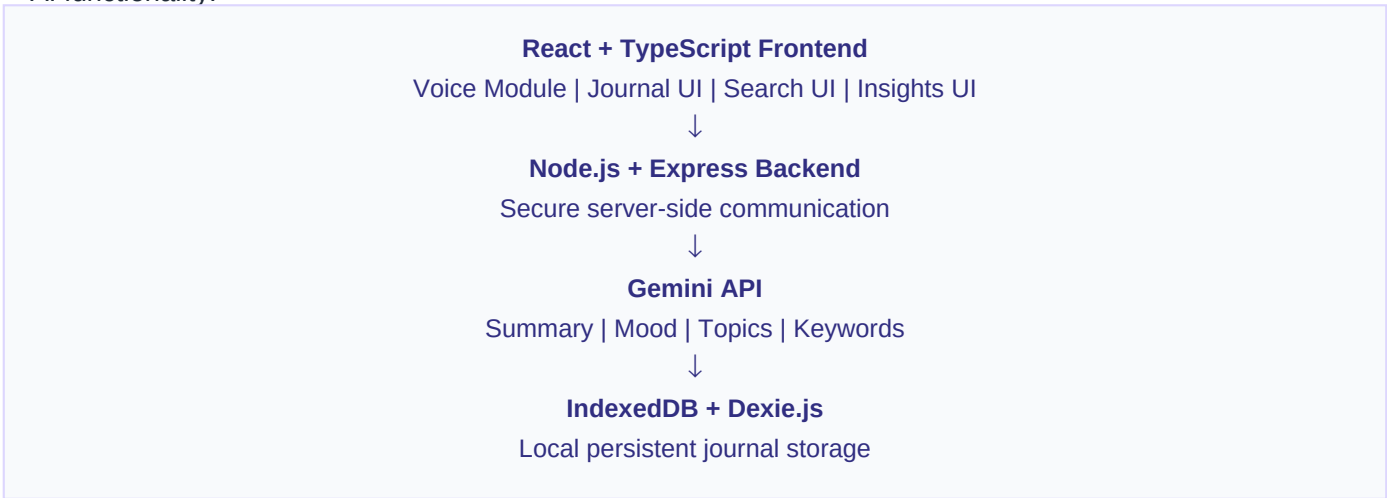
- 1 **Open Sentora:** The user accesses the journaling interface.
- 2 **Record thoughts:** The user speaks naturally using the microphone.
- 3 **Convert speech to text:** Speech recognition produces a transcript.
- 4 **Review and edit:** The user can correct transcription errors before analysis.
- 5 **Request AI analysis:** The user explicitly chooses to analyze the entry.
- 6 **Generate structured results:** AI returns a summary, emotional tone, topics, and keywords.

- 7 **Save locally:** The entry and associated metadata are stored in the browser.
- 8 **Rediscover later:** Timeline, smart search, and insights help retrieve memories.



5. System Architecture

Sentora uses a practical architecture designed to keep the project achievable while still delivering meaningful AI functionality.



Important design principle: AI is used where it adds value. Normal application tasks such as saving data, sorting entries, rendering charts, and navigating pages are handled using standard programming rather than unnecessarily sending everything to an AI model.

6. Technology Stack

Technology	Role in Sentora
React	Builds the interactive frontend and user interface.
TypeScript	Adds type safety and structure to the application code.
Vite	Provides the frontend development and build environment.
Tailwind CSS	Creates responsive layouts and consistent visual styling.

Technology	Role in Sentora
Node.js	Runs JavaScript on the server side.
Express	Creates backend endpoints and handles communication with Gemini.
Google Gemini API	Performs AI-powered text analysis.
MediaRecorder API	Captures microphone audio in the browser.
Web Speech API	Converts speech into text where browser support is available.
IndexedDB	Stores journal data locally in the browser.
Dexie.js	Simplifies IndexedDB database operations.
Recharts	Creates analytics and visualization components.
Web Crypto API	Can support optional local protection features.

7. AI-Powered Journal Analysis

Gemini is used as the intelligence layer for the journal text. The project does not claim to train its own large language model. Instead, it integrates an existing AI capability into a complete application designed around a real user problem.

A. Summary generation

Long or unstructured journal text can be condensed into a concise description of the main events or ideas.

B. Emotional tone / mood classification

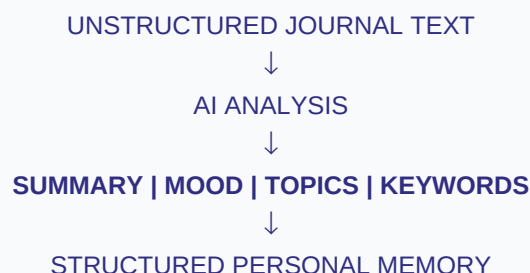
The system identifies emotional language expressed in the text. This feature is for reflection and organization only and must not be presented as medical, psychiatric, or psychological diagnosis.

C. Topic extraction

The AI identifies broad subjects discussed in an entry, such as education, career, software development, relationships, or travel.

D. Keyword extraction

Important concepts and terms are extracted to improve organization and later retrieval.

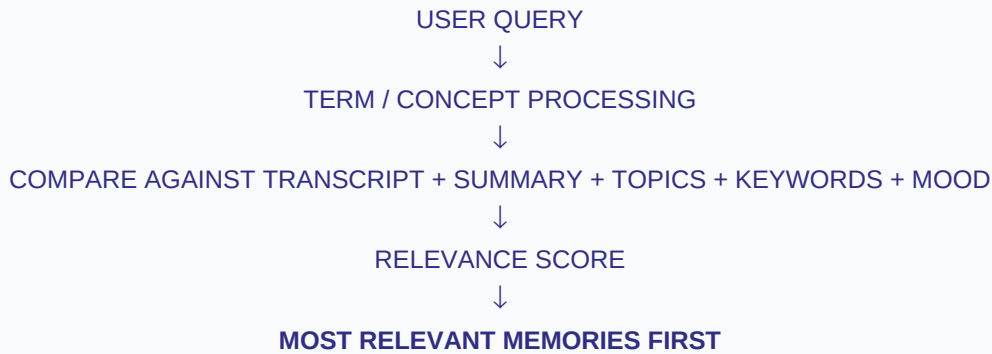


8. Smart Memory Search

The purpose of smart search is to help users retrieve an experience even when they do not remember the exact words used in the original journal.

- **Transcript match:** Search the original journal text.

- **Summary match:** Search the AI-generated condensed representation.
- **Topic match:** Compare the query with extracted subjects.
- **Keyword match:** Compare important concepts.
- **Emotion match:** Use emotional metadata where relevant.
- **Synonym and related-term support:** Expand queries with simple related terms where implemented.
- **Relevance scoring:** Rank entries based on how strongly they match.

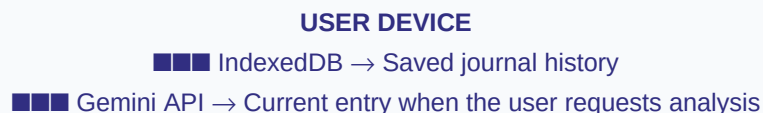


9. Data Storage and Privacy Approach

Sentora follows a **local-first** approach. Saved journal entries are primarily stored in the user's browser using IndexedDB.

- Journal transcript
- AI-generated summary
- Mood or emotional tone information
- Topics
- Keywords
- Dates and activity metadata
- Audio recordings where the user chooses to keep them

The privacy claim must be explained honestly. The complete journal history is not automatically sent to Gemini. When the user explicitly requests AI analysis, the current selected journal text is sent to the AI service for processing.



A stronger security implementation can later use the Web Crypto API for local protection. However, the project should not claim end-to-end encryption or other security guarantees unless those features are actually implemented and verified.

10. Example Journal Analysis

Example input:

“Today I was really stressed because my project deadline is close. I spent most of the day fixing bugs, and at one point I thought I would not finish. But after working with my team, we solved the main issue. I feel much better now.”

Analysis Type	Example Output
Summary	The user experienced stress about an approaching project deadline but felt relieved after resolving an im
Primary emotional tone	Stress / anxiety, followed by relief and improved confidence.
Topics	Education, project work, software development, teamwork.
Keywords	Project, deadline, bugs, team, stress, solution.

11. Real-World Problem Cases and Use Cases

1. University students

Students can quickly reflect on exams, assignments, projects, interviews, teamwork, or daily experiences without writing long entries. Later they can search for memories related to a particular viva, deadline, or learning experience.

2. Personal memory preservation

Important life events such as trips, birthdays, achievements, first days, or meaningful conversations can be captured naturally and rediscovered years later.

3. Busy professionals

A professional may have a few minutes to speak but not enough time to write a detailed journal entry. Voice-first interaction reduces the effort of capture.

4. Idea and project capture

Users often have useful ideas while travelling, walking, or between activities. Sentora can capture the thought and organize it into searchable information.

5. Personal reflection

Topic frequency, journaling activity, and emotional language patterns can help users reflect on what they have been discussing over time, without positioning the system as a medical assessment tool.

6. Long-term personal archive

Over time, Sentora can function as a searchable timeline of life experiences. Example searches include: “My first college project”, “When did I start learning Java?”, or “Times I talked about my career.”

12. Why AI Is Necessary

A normal voice journal can capture and store text. AI adds value by organizing and interpreting the meaning of unstructured content.

WITHOUT AI

Voice → Text → Stored

WITH SENTORA'S AI LAYER

Voice → Text → Summary + Mood + Topics + Keywords → Structured Memory

A strong pitch explanation is:

“Voice recording alone captures information, but it does not organize or understand it. AI adds value by converting unstructured journal text into structured information such as summaries, emotional tone, topics, and keywords. This improves organization, search, and personal insights.”

13. Why Use Gemini Instead of Building an AI Model

Training a large language model from scratch requires large datasets, significant computational resources, specialized infrastructure, and substantial machine-learning expertise. The purpose of Sentora is to build an AI-powered journaling application, not to train a foundational language model.

Therefore, Gemini acts as an AI service inside the broader application architecture.

SENTORA'S INNOVATION



Not: “We built a new large language model”



Instead: “We designed a practical system that integrates AI capabilities to solve a personal journaling and memory-retrieval problem.”

14. What Makes Sentora Innovative

- **Voice-first interaction:** Lowers the barrier to creating journal entries.
- **User-controlled AI analysis:** The user decides when an entry is analyzed.
- **Structured memory creation:** AI converts free-form text into searchable metadata.
- **Local-first storage:** Saved history is primarily stored on the device.
- **Memory-oriented search:** The system aims to retrieve experiences, not only exact phrases.
- **Personal insights:** Data accumulated over time can be visualized for reflection.

The innovation is primarily in the combination and integration of these capabilities into one coherent personal memory experience.

15. Best Pitch Statements

One-minute pitch

“Sentora is a privacy-first, voice-first AI journaling application designed to make personal journaling easier and more intelligent. Instead of requiring users to type long journal entries, Sentora allows them to naturally speak their thoughts. The application converts speech into editable text and uses Gemini AI to generate a summary, identify emotional tone, extract topics, and identify important keywords. The processed journal entry is stored locally in the user’s browser using IndexedDB. Over time, Sentora helps users rediscover their personal memories through smart search, timelines, and visual insights. The goal is to transform traditional journaling from a collection of text entries into a searchable and organized personal memory system.”

Technical pitch

“Sentora uses a React and TypeScript frontend with browser-native MediaRecorder and Web Speech APIs for voice interaction. A Node.js and Express backend communicates with the Gemini API for structured journal analysis. Gemini generates summaries, emotional tone classifications, topics, and keywords. Journal entries and associated data are stored locally using IndexedDB, managed through Dexie.js. The application implements hybrid local search across transcripts, summaries, topics, keywords, and emotional metadata. Recharts is used to visualize personal journaling patterns and trends.”

16. Viva and Pitch Questions You Should Be Ready For

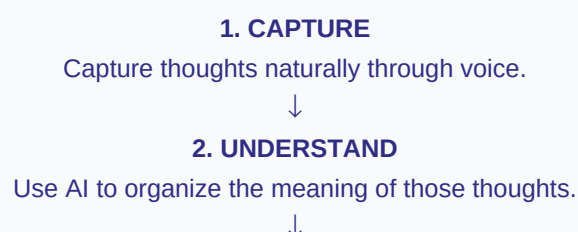
Question	Key Answer
Why is AI needed?	AI converts unstructured journal text into summaries, emotional tone information, topics, and keywords.
Why Gemini?	The project focuses on application design and AI integration rather than training a foundational model.
Is this a mental-health diagnosis system?	No. Mood analysis reflects emotional language in the journal and is not a medical or psychological diagnosis.
Why IndexedDB?	It supports local persistent storage without requiring a paid cloud database for the prototype.
What does privacy-first mean here?	Saved journal history is primarily stored locally. The current journal text is sent to Gemini only when needed for analysis.
Why not just build a voice recorder?	Recording captures information but does not organize it. AI creates structured metadata that improves searchability.
What is the main innovation?	The combination of voice-first capture, AI understanding, local-first storage, and intelligent memory recall.

17. Future Scope

- **Multilingual journaling:** Support additional languages for voice and text.
- **Improved semantic search:** Add embedding-based similarity when the project grows.
- **Offline speech-to-text:** Reduce dependence on browser speech recognition.
- **Mobile application:** Extend the experience to Android and iOS.
- **Optional encrypted backup:** Allow users to back up selected data securely.
- **Multi-device synchronization:** Add opt-in synchronization without changing the local-first philosophy.
- **Custom journaling prompts:** Encourage consistent reflection.
- **Calendar integration:** Connect memories and activity with dates and events.

18. Final Positioning

The simplest and strongest way to understand Sentora is through three actions:



3. REDISCOVER

Find meaningful memories through search, timelines, and insights.

Final positioning statement:

“Sentora is a voice-powered personal memory system that captures unstructured thoughts, uses AI to organize their meaning, and helps users rediscover important moments over time.”

This document describes the current practical prototype architecture. Any presentation or viva claim should remain consistent with the features actually implemented and tested.